

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

AStSD9BU3X3YDhVwiuB8eTuTmRLCyM45eZzL3vFr1wZT

Date: 5/1/2025

Compounds: Amoxicillin, Aspirin, Paracetamol

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Amoxiparspirin

Formula: C₁₉H₂₂N₂O₈S

Weight: 438.46

Drug-likeness: 0.74

Activity:

Antibacterial, Anti-inflammatory, Analgesic

Target Analysis

Penicillin-binding proteins (PBPs)

Binding Affinity: -9.7

Essential in bacterial cell wall synthesis inhibition

Cyclooxygenase enzymes (COX-1 and COX-2)

Binding Affinity: -8.5

Reduction of inflammation and pain

Toxicity Profile

Risk Level: Moderate

Affected Systems: Liver, Stomach, Kidneys

Mitigation Suggestions:

- Use lowest effective dose
- Monitor liver enzyme levels
- Supplement with protective agents like proton pump inhibitors

Pharmacokinetics

Absorption:

High oral absorption

Distribution:

Widely distributed across tissues, varies per component

Metabolism:

Hepatic, via cytochrome P450 enzymes

Excretion:

Renal elimination of metabolites

Half-life: Amoxicillin: 1 hour, Aspirin: 3 hours, Paracetamol: 2-3 hours

Detailed Analysis

Amoxiparspirin combines features from Amoxicillin, Aspirin, and Paracetamol to create a drug intended to treat bacterial infections while providing simultaneous anti-inflammatory and analgesic benefits. This compound leverages the antibacterial capabilities of amoxicillin through inhibition of bacterial cell wall synthesis. Concurrently, aspirin and paracetamol components alleviate inflammatory responses and pain by targeting COX enzymes and modulating prostaglandin synthesis. The dual activity can prove beneficial in multi-symptomatic treatment settings, such as bacterial infections accompanied by significant inflammation and pain. Predicted moderate toxicity primarily affects the liver and stomach. Pharmacokinetic parameters suggest good absorption and wide tissue distribution, desirable for systemic effects. Initial studies highlight a requirement for meticulous monitoring due to possible adverse reactions. This makes Amoxiparspirin a significant candidate for further pharmaceutical exploration aiming at refined, multi-functional therapeutics.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>